Lab 06 – MATH 240 – Computational Statistics

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Abstract

In this lab we used an API to extract US Census data in New Haven County, CT. We then compared various renter data to home owner data. This could be useful to compare how income and expenses changed over time for renters and owners.

Keywords: Application Programming Interface: Enables applications to share data, features, and functionality;

1 Introduction

The goal of this lab is to analyze data from the US Census API, about New Haven County owners and renters to evaluate cost of living.

2 Methods

To analyze renter and homeowner costs in New Haven County, CT, we used the `tidycensus` package (Walker and Herman, 2025) which allowed us to get ACS data from the US Census API. Specifically we wanted :

- Total Population (B02001_001E)
- Population that identifies as Black or African American Alone (B02001_003E)
- Median Income (B25119_001E)
- Median Income for Owners (B25119_002E)
- Median Income for Renters (B25119_003E)
- Median Monthly Owner Costs (B25088_002E)
- Median Monthly Owner Costs as a Percentage of Household Income (B25092_002E)
- Median Gross Rent (B25031_001E)
- Median Gross Rent as a Percentage of Household Income (B25071_001E)

2.1 Data Processing

Each year's ACS data were reshaped using `pivot_wider()` from (Wickham et al., 2019) to arrange variables into a more accessible format. The data sets from 2013 to 2023 were then merged into a single data set. Finally, tract identifiers were standardized by mapping 2022 census tracts to 2020 equivalents where necessary. Additionally, tracts with insufficient data across multiple years were filtered out.

2.2 Graph Creation

The first comparison we made was comparing monthly owner costs as a percentage of income and monthly rent as a percentage of income geographically. Using the `sf` (Pebesma, 2018) and `tigris` (Walker, 2024) packages, I created a geographical data set for 2023 and used the given code to get the map and town names of New Haven County. Combining these I created a heat map for both monthly owner costs as a percentage of income and monthly rent as a percentage of income.

Using the data we had, the best comparison we could make was with median monthly costs/rent, monthly costs/rent as a percentage of income, and median income of homeowners and renters. Using `ggplot2` (Wickham, 2016), I first created a line plot by tract comparing the renter and homeowner data (renter and home owner data being set to different colors) for each of these variables. It was difficult to see trends with the line plots because there were so many tracts, so i used `geom_smooth()`, also from `ggplot2`, to create a trend line for both renter and homeowner data.

Next, to compare not only the average median gross rent of New Haven County, but also the spread of data by tract, I used `ggridges` (Wilke, 2024) to create a ridge plot. The ridge plot has an advantage over the line plots or heat maps because it shows the spread of data for each year clearly.

3 Results

3.1 Geographical Analysis

The geographical heat maps I created (Figure 1 and Figure 2) show a stark difference between monthly owner costs as a percentage of income and monthly rent as a percentage of income. In nearly every single town the rent as a percentage of income is much more than monthly owner costs.

3.2 Owner and Renter Trends over Time

Gross Expenses/Rent: Owner expenses and rent both follow very similar trends over time as seen in Figure 3, but gross owner expenses are on average around $1,000 more than rent.

Expenses/Rent as a Percent: From Figure 4 we can see that rent as a percent of income stays relatively the same from 2013-2023, only decreasing a little, but owner cost as a percent of income decreases by around 5% from 2013-2023. Additionally, it appears that renter data has much more variability than the owner data, shown by the spread of the line plots. The trend line for renters is always above that of owners, around 5% in 2013 and around 8% in 2023.

Income: From Figure 5 we can see clearly that median income of owners increases at a faster rate than median income of owners from 2013-2023. We can also see that the trend line for owners income is around $50,000 greater than the renters in 2013 and around $100,000 greater in 2023. I think it is also worthy to note that income of owners has a higher variability than renters, shown by the spread of the line plots.

3.3 Ridge Plot of Median Gross Rent

The ridge plot of median gross rent (Figure 6) shows a right skewed distribution from 2013 to 2020 with slight increases in the skewness. After 2020, however, the distribution begins to shift from a right skewed distribution to an almost bi-modal distribution by 2023. In addition to showing the variability over time, the ridge plot also shows an increase of the mean over time, going from around $1,100 to around $1,500.

4 Discussion

While the purpose of this lab was to explore how cost of housing may vary for those who choose to rent and those who can and choose to own, the results tell an interesting story about the difference between renters and owners.

4.1 Owner and Renter trends over time

The results of the gross cost graph tell us that on average it is more expensive to be a home owner than a renter.

The results of the cost as a percent of income graph tell us a little more than the previous one. It shows that on average owners pay less and less as a percentage of income every year while renters only see a small decrease in rent as a percent of income. This is very different than what the median gross cost graph shows, which is the same trend for both renters and owners. This implies that owners have a greater increase in income over time than renters.

The results of the income graph show exactly this: a major increase in owner income and minor increase in renter income. This data shows us what is happening but not why.

4.2 Why are these Trends Happening

The density ridge plot gives a little more context. With the increase in rent over time, we see a widening spread in the distribution of rental costs. This suggests that while some renters continue to pay relatively low rent, others face significantly higher costs, leading to greater inequality among renters. This could be due to rising demand in certain areas, inflationary pressures, or policy changes affecting housing affordability.

One possible explanation for the divergence in income trends is that homeownership is often associated with wealth accumulation. Homeowners benefit from property appreciation, tax incentives, and stable mortgage payments, which may contribute to their growing financial stability over time. Renters, on the other hand, are more vulnerable to rising rental costs, and wage stagnation, which could explain the relatively slow increase in their incomes.

References

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5 Appendix

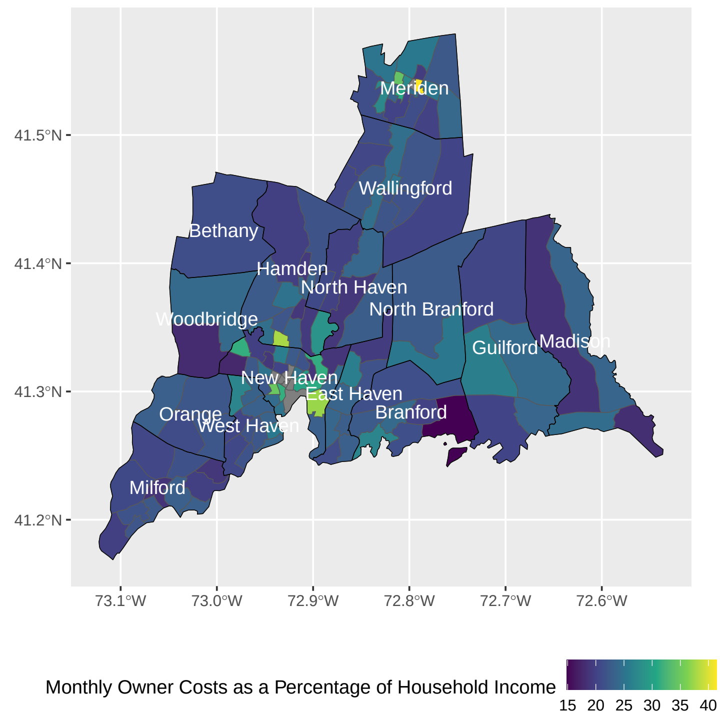

Figure 1: Map of New Haven County comparing median monthly owner costs as a percentage of household income by census tract in 2023

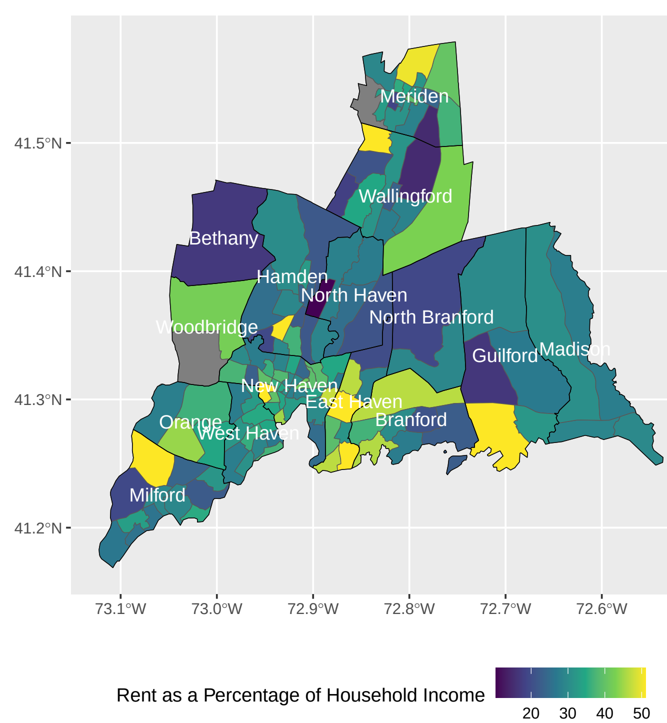

Figure 2: Map of New Haven County comparing median rent as a percentage of household income by census tract in 2023

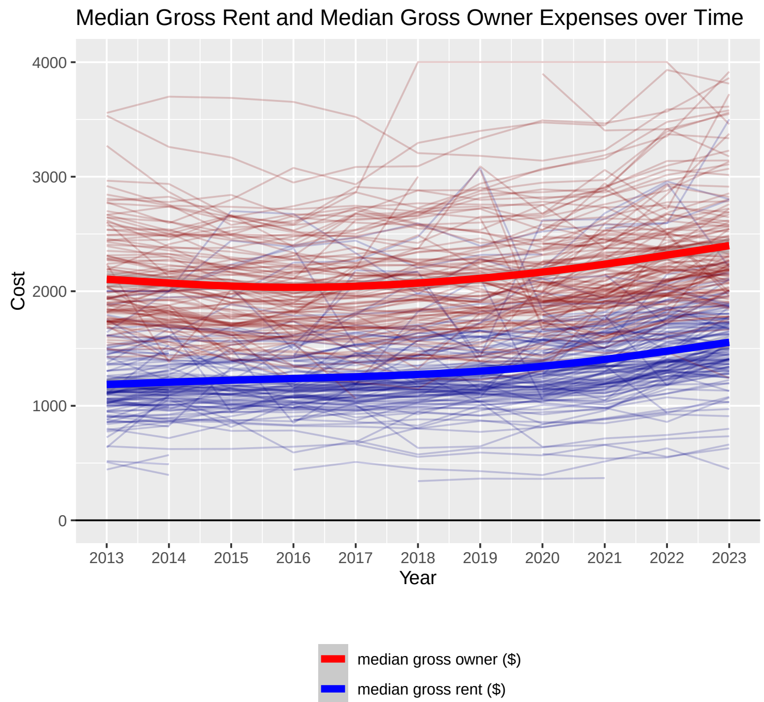

Figure 3: Line plot of median gross owner expenses and median gross rent by tract from 2013-2023 (with lines of best fit)

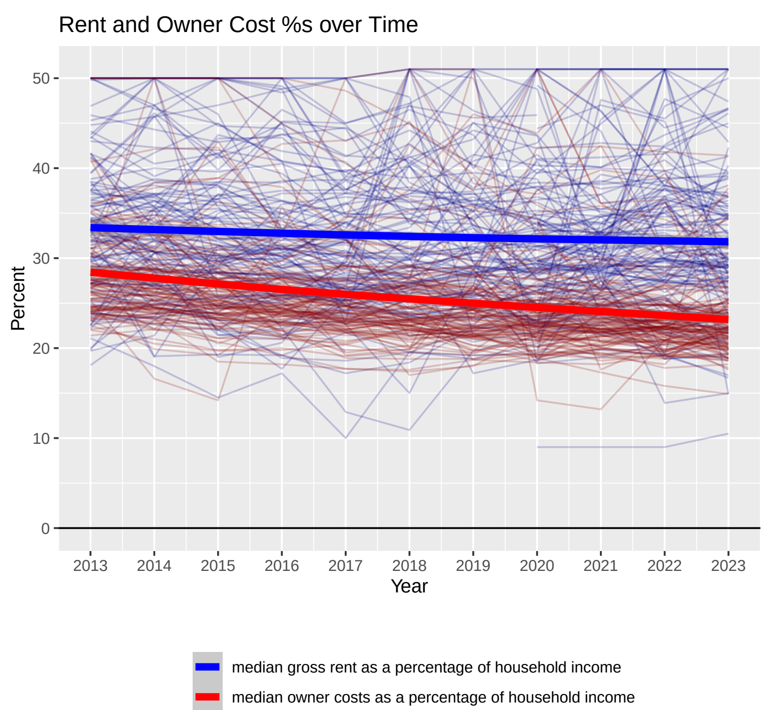

Figure 4: Line plot of median owner expenses and median rent as a percent of income by tract from 2013-2023 (with lines of best fit)

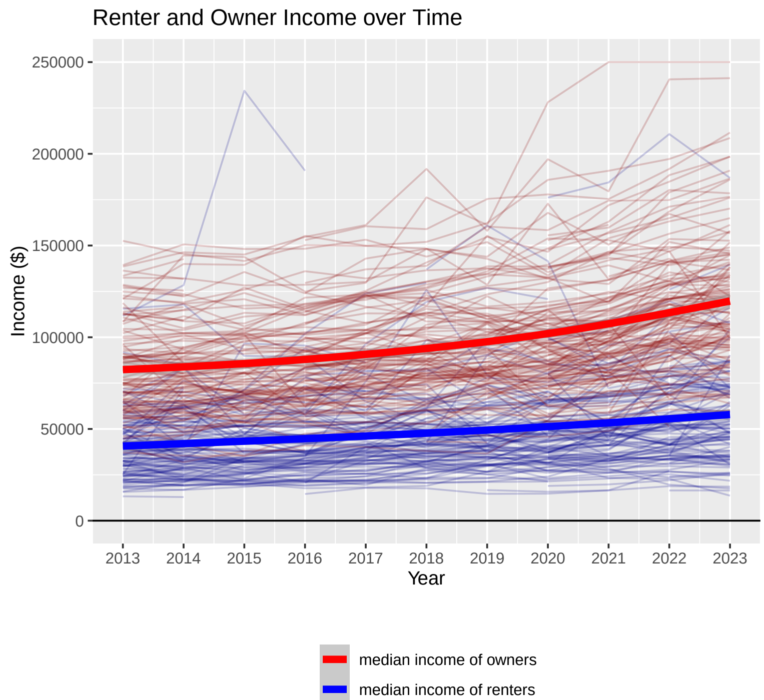

Figure 5: Line plot of median owner income and median renter income by tract from 2013-2023 (with lines of best fit)

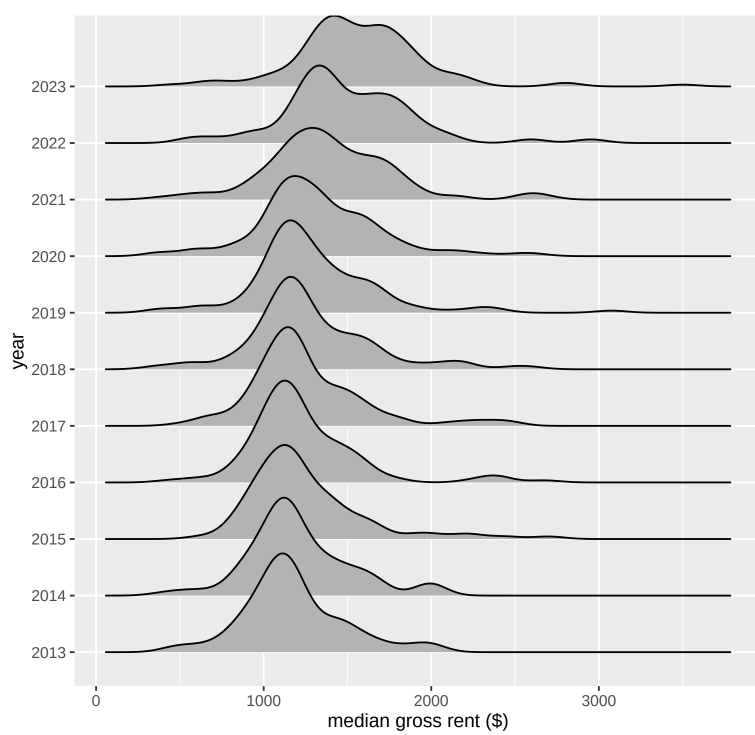

Figure 6: Density Ridge plot of median gross rent over time